

Full-stack Development

ECTS

5

Prerequisites

WEB2 or similar.

Main purpose

The purpose of this code is to learn to design and implement full stack web applications.

Knowledge

The student should be able to

- explain the elements of the TypeScript type system.
- explain the function of TypeScript utility types.
- explain the definition of functional programming.
- explain relevant client programming design patterns.
- explain mechanisms for rendering and re-rendering.
- explain the components of reactive programming.

Skills

The student should be able to

- apply object-oriented programming in TypeScript.
- apply functional programming in TypeScript.
- apply callbacks and higher-order functions in TypeScript.
- apply at least two web client frameworks and at least two state management frameworks.
- implement a server using TypeScript.
- implement server-side rendering.
- use reactive programming in the client.
- argue for the choice of server-side and client-side rendering.
- argue for the choice of state management techniques in web client.

Competences

At the end of the course, the students should be able to

- design and implement a web application using one or more of the techniques and technologies taught in the course.

Topics

TypeScript, Client programming, Server programming, Server-side rendering, Functional programming, Reactive programming

Teaching methods and study activities

Lessons alternate between theory and practical exercises.

The course contains one or more assignments. The assignments must be handed in by the given deadline, or the student will be denied attending the exam and will thus have used one exam attempt.

Expected workload for students is estimated to 135 hours.

Exam

Exam prerequisites:

None

Type of exam:

Individual oral exam, 20 minutes, without preparation.

Exam depends on 2 of the 6 course assignments, handed in according to deadline.

Exam covers the assignments and the part of the syllabus relevant to the assignments.

The final grade will be based on an overall assessment of the 6 assignments and the oral examination.
Internal assessment

Tools allowed:
All

Re-exam:
Conducted as the ordinary exam.

Grading criteria

Grading based on the Danish 7-point scale.

Additional information Responsible

Ole I. Hougaard (oih)

Valid from

01-08-2024

Course type

Elective for Software Engineering. Elective for the specialization Interactive Media
Valgfag for Softwareingeniør. Valgfag for Interactive Media specialisering.